HW08

due on March 12, Friday.

P1 **Problem** [redacted]. Use a computer to sum the rotational partition function (equation [redacted]) algebraically, keeping terms through $\ell = 6$. Then calculate the average energy and the heat capacity. Plot the heat capacity for values of kT/ϵ ranging from 0 to 3. Have you kept enough terms in Z to give accurate results within this temperature range? (1)

[20pts]

$$Z = \sum_{\ell=0}^{\infty} (2\ell+1)\, e^{-\beta\varepsilon_0 \ell(\ell+1)} \qquad (1) \quad \text{for } \frac{1}{\beta\varepsilon_0} \in [0,3].$$

P2 **Problem** [redacted]. In this problem you will investigate the behavior of ordinary hydrogen, H_2, at low temperatures. The constant ϵ is 0.0076 eV. As noted in the text, only half of the terms in the rotational partition function, equation 6.30, contribute for any given molecule. More precisely, the set of allowed ℓ values is determined by the *spin* configuration of the two atomic nuclei. There are four independent spin configurations, classified as a single "singlet" state and three "triplet" states. The time required for a molecule to convert between the singlet and triplet configurations is ordinarily quite long, so the properties of the two types of molecules can be studied independently. The singlet molecules are known as **parahydrogen** while the triplet molecules are known as **orthohydrogen**.

(20pts) (a) For parahydrogen, only the rotational states with <u>even</u> values of ℓ are allowed.* Use a computer (as in Problem [redacted] P1) to calculate the rotational partition function, average energy, and heat capacity of a parahydrogen molecule. Plot the heat capacity as a function of kT/ϵ.*

(20pts) (b) For orthohydrogen, only the rotational states with <u>odd</u> values of ℓ are allowed. Repeat part (a) for orthohydrogen.

(20pts) (c) At high temperature, where the number of accessible even-ℓ states is essentially the same as the number of accessible odd-j states, a sample of hydrogen gas will ordinarily consist of a mixture of 1/4 parahydrogen and 3/4 orthohydrogen. A mixture with these proportions is called **normal hydrogen**. Suppose that normal hydrogen is cooled to low temperature without allowing the spin configurations of the molecules to change. Plot the rotational heat capacity of this mixture as a function of temperature. At what temperature does the rotational heat capacity fall to half its high-temperature value (i.e., to $k/2$ per molecule)?

(20pts) (d) Suppose now that some hydrogen is cooled in the presence of a catalyst that allows the nuclear spins to frequently change alignment. In this case *all* terms in the original partition function are allowed, but the odd-ℓ terms should be counted three times each because of the nuclear spin degeneracy. Calculate the rotational partition function, average energy, and heat capacity of this system, and plot the heat capacity as a function of kT/ϵ.

(e) ~~A deuterium molecule, D_2, has nine independent nuclear spin configurations, of which six are "symmetric" and three are "antisymmetric." The rule for nomenclature is that the variety with more independent states gets called "ortho-," while the other gets called "para-." For orthodeuterium only even-j rotational states are allowed, while for paradeuterium only odd-j states are allowed.† Suppose, then, that a sample of D_2 gas, consisting of a normal equilibrium mixture of 2/3 ortho and 1/3 para, is cooled without allowing the nuclear spin configurations to change. Calculate and plot the rotational heat capacity of this system as a function of temperature.‡~~